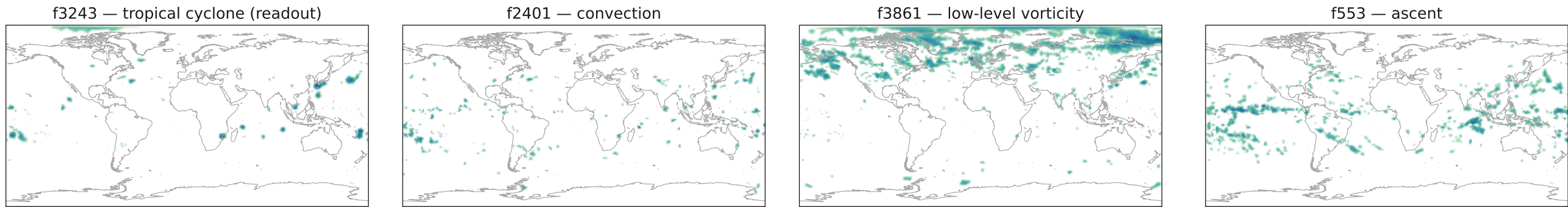
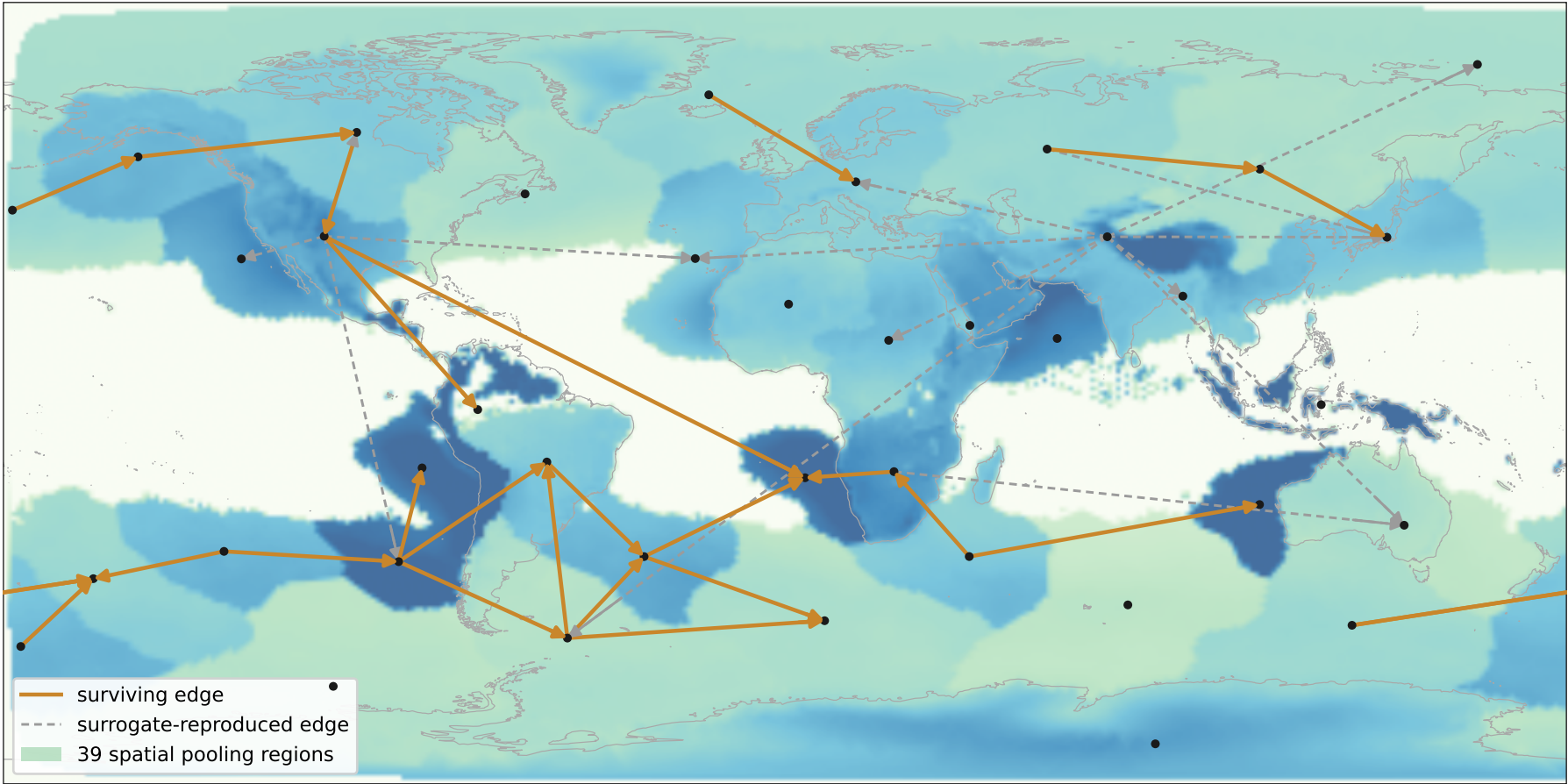


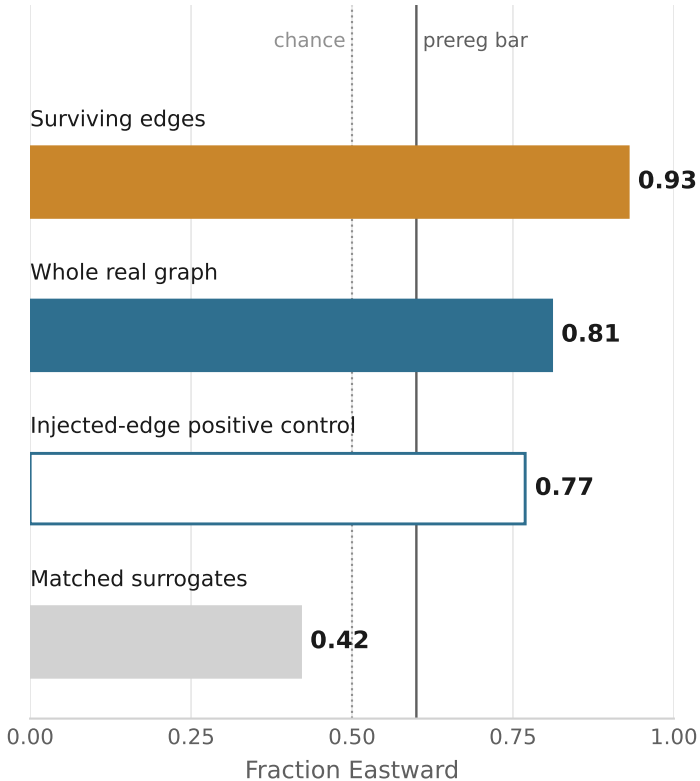
a SAE activation maps — layer 8, accumulated over 12 dates (darker = fires more often there)



b Inferred lagged edges on the map — 39-region spatial partition, 38 consensus edges (source → target centroid)



c Real vs matched surrogate: eastward asymmetry lives in the surviving edges



Fraction of scoreable extratropical zonal pairs whose arrow points east. Surrogates match spectrum, sparsity, footprint overlap and same-time covariance (S4c); only lead-lag asymmetry is left, and it points east.